

Here's a well-rounded practice lab that combines **EtherChannel** with the topics you've already practiced (VLSM + SVI + VLANs + Trunking):

EtherChannel Practice Lab – Stellar Enterprises

Main Network: 172.16.40.0/21 (2048 addresses)

You are required to design the network using **VLSM** and implement **Layer 2 EtherChannel** for better bandwidth and redundancy.

Departments:

VLAN ID	Department	Hosts Needed
10	Management	90 hosts
20	Sales	140 hosts
30	Engineering	60 hosts

Devices:

- 1 × Layer 3 Switch (3560 or 3750) — will do Inter-VLAN routing using SVI
 - 2 × Layer 2 Switches (2960)
 - 25 × PCs
-

Requirements:

1. VLSM Design

Design an efficient subnet scheme for the 3 departments from 172.16.40.0/21 .

2. VLAN Configuration

- Create VLAN 10 (Management), VLAN 20 (Sales), VLAN 30 (Engineering)

3. EtherChannel (Main Focus)

- Configure **Layer 2 EtherChannel** between the two Layer 2 switches using **2 physical links** (use `GigabitEthernet` ports).
- Use **LACP** (Link Aggregation Control Protocol) as the negotiation protocol.

4. Trunking & SVI

- Configure trunks from both Layer 2 switches to the Layer 3 switch.
- Configure **SVI** on the Layer 3 switch for all 3 VLANs (Inter-VLAN routing).

5. Verification

- All PCs in the same VLAN can ping each other.
 - PCs in different VLANs can ping each other.
 - Check EtherChannel is working (`show etherchannel summary`).
-

Bonus Challenge:

Configure the EtherChannel with **static mode** (`mode on`) on one side and **LACP active** on the other side, then fix it so it works properly.

Would you like me to:

- Give you a **blank VLSM table** first?
- Provide the **recommended VLSM solution**?
- Or give you the full configuration guide after you finish designing?

Just tell me how you want to proceed!